

An explicit formula for a weight enumerator of linear-congruence codes

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An explicit formula for a weight enumerator of linear-congruence codes is provided. This extends the work of Bibak and Milenkovic [IEEE ISIT (2018) 431–435] addressing the binary case to the non-binary case. Furthermore, the extension simplifies their proof and provides a complete solution to a problem posed by them.

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Introduction

Throughout this article, n and m denote positive integers, b denotes an integer and $\mathbb{Z}_q := \{0, 1, \dots, q-1\} \subset \mathbb{Z}$ for a positive integer q . We will use n for a code length, m for a modulus, b for a defining parameter of a code and $\mathbb{Z}_q$ for a code alphabet.

Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{Z}^n$ and $b \in \mathbb{Z}$. The set C of all the solutions $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{Z}_q^n$ for a linear congruence equation $\mathbf{a} \cdot \mathbf{x} \equiv b \pmod{m}$ is said to be a *linear-congruence code* where $\mathbf{a} \cdot \mathbf{x} := a_1x_1 + \dots + a_nx_n$. A linear-congruence code C is called *binary* when $q = 2$.

Several deletion-correcting codes which have been studied are linear-congruence codes; the Varshamov-Tenengol'ts codes, the Levenshtein codes, the Helberg codes, the Le-Nguyen codes, the construction C' of Hagiwara (for some parameters), the consecutively systematic encodable codes and the ternary integer codes in fall into this category (Table).

Examples of linear-congruence codes

Linear-congruence code	q	$\mathbf{a} = (a_1, \dots, a_n)$	m	Constraints
Varshamov-Tenengol'ts code	2	$(1, \dots, n)$	$n+1$	
Levenshtein code	2	$(1, \dots, n)$	m	$m \geq n+1$
Helberg code	2	$(v_1, \dots, v_n)$	v_{n+1}	$s \in \mathbb{Z}_{>0}$
Le-Nguyen code	q	$(w_1, \dots, w_n)$	m	$m \geq w_{n+1}, s \in \mathbb{Z}_{>0}$

Construction C'	2	$(c_1, \dots, c_n)$	$b \not\equiv 0, n(n+1)/2 \pmod n$
Consecutively systematic encodable codes	2	$(b_1, \dots, b_n)$	$b = 0, s \in \mathbb{Z}_{>0}, 0 < n - s < 2^s + 1$ $2^s - 1$
Ternary integer code	3	$(t_1, \dots, t_n)$	$2^{n+1} - 1$

The following problem concerning the size of a linear-congruence code—the number of solutions for a linear congruence equation [\[eq: \$ax = b\$ \]](#)—is posed by Bibak and Milenkovic.

Give an explicit formula for the size of a linear-congruence code.

Finding an explicit formula would be a first step toward understanding the asymptotic behavior of the size of a linear-congruence code. Bibak and Milenkovic provide a solution to the problem for the binary case. In this article, we provide a complete solution to the problem with a simple proof, which improves the argument of Bibak and Milenkovic. Actually, what we will show is how the Hamming weights of the solutions for a linear congruence equation distribute. This immediately gives an expression of the size of a linear-congruence code involving exponential sums—Weyl sums of degree one.

To state the main theorem we need notation which will be standard.

For a code $C \subseteq \mathbb{Z}_q^n$, we define a polynomial $W_C(z)$ by

$$W_C(z) = \sum_{x \in C} z^{\text{wt}(x)} = \sum_{i=0}^n A_i(C) z^i,$$

where $\text{wt}(x)$ denotes the Hamming weight and

$$A_i(C) = |\{x \in C : \text{wt}(x) = i\}| \quad (0 \leq i \leq n).$$

The polynomial $W_C(z)$ is said to be the (non-homogeneous) *weight enumerator* of the code C .

Following custom due to Vinogradov in additive number theory, $e(\alpha)$ denotes

$$e^{2\pi i \alpha \sqrt{-1}}$$

for $\alpha \in \mathbb{R}$. Now we are in position to state our main theorem.

Let $a = (a_1, \dots, a_n) \in \mathbb{Z}^n$ and $b \in \mathbb{Z}$. Then the weight enumerator $W_C(z)$ of the linear-congruence code

$$C = \{x \in \mathbb{Z}_q^n : a \cdot x \equiv b \pmod m\}$$

is given by

$$W_C(z) = \frac{1}{m} \sum_{j=1}^m e\left(-\frac{jb}{m}\right) \prod_{i=1}^n \left(1 + z e\left(\frac{ja_i}{m}\right) + \dots + z e\left(\frac{ja_i(q-1)}{m}\right)\right).$$

With the same notation as above, the size of the code C is given by

$$|C| = \frac{1}{m} \sum_{j=1}^m e\left(-\frac{jb}{m}\right) \prod_{i=1}^n \left(1 + e\left(\frac{ja_i}{m}\right) + \dots + e\left(\frac{ja_i(q-1)}{m}\right)\right).$$

Proof of Theorem

The only lemma we need to prove the main theorem is the following trivial one.

$$\frac{1}{m} \sum_{j=1}^m e\left(\frac{jb}{m}\right) = \begin{cases} 1 & \text{if } b \equiv 0 \pmod m \\ 0 & \text{if } b \not\equiv 0 \pmod m \end{cases}$$